1. **Write a shell script to calculate the sum of two numbers.**

```
echo "Enter first number: "
read a
echo "Enter second number: "
read b
sum=`expr $a + $b`
echo "Sum = $sum"
```

2. **Write a shell script to find the modulus (remainder) of two numbers**

```
echo "Enter dividend: "
read a
echo "Enter divisor: "
read b
mod=`expr $a % $b`
echo "Remainder = $mod"
```

3. **Write a shell script to accept marks of three subjects, calculate the total marks and percentage, and display the result.**

```
echo -n "Enter marks of Subject 1: "
read m1

echo -n "Enter marks of Subject 2: "
read m2
```

```
echo -n "Enter marks of Subject 3: "
read m3

total=`expr $m1 + $m2 + $m3`
percentage=`expr $total / 3`


echo "--------------------------------"
echo "Subject 1 : $m1"
echo "Subject 2 : $m2"
echo "Subject 3 : $m3"
echo "--------------------------------"
echo "Total Marks : $total"
echo "Percentage  : $percentage%"
```

4. **Write a shell script to accept two numbers and perform addition, subtraction, multiplication, and division using the expr command.**

```
echo "Enter first number: "
read n1

echo "Enter second number: "
read n2

sum=`expr $n1 + $n2`
sub=`expr $n1 - $n2`
mul=`expr $n1 \* $n2`
div=`expr $n1 / $n2`

echo "Addition       : $sum"
echo "Subtraction    : $sub"
echo "Multiplication  : $mul"
echo "Division       : $div"
```

5 . Write a shell script to accept the basic salary from the user and calculate DA (10%), HRA (20%), and gross salary using the expr command.

Gross salary = Basic + DA + HRA

```
echo -n "Enter your basic salary: "
read salary

da=`expr $salary \* 10 / 100`
hra=`expr $salary \* 20 / 100`

#
gross=`expr $salary + $da + $hra`

echo "Basic Salary : $salary"
echo "Dearness Allowance (10%) : $da"
echo "House Rent Allowance (20%) : $hra"
echo "Gross Salary : $gross"
```

5. Shell Script: Special Variables Example

```
echo "Script name: $0"
echo "First argument: $1"
echo "Second argument: $2"
echo "Total number of arguments: $#"
echo "Process ID of current shell: $$"
echo "Exit status of last command: $?"
```

7.Shell Script: Environment Variables Example

```
echo "PATH variable: $PATH"
echo "Home directory: $HOME"
echo "Logged-in user: $USER"
echo "Default shell: $SHELL"
echo "Present working directory: $PWD"
echo "Mailbox location: $MAIL"
echo "Primary prompt string (PS1): $PS1"
echo "Secondary prompt string (PS2): $PS2"
```

8. create script where you pass two numbers as $1 and $2, and then it shows all operations (addition, subtraction, multiplication, division, modulus) in one go.

```
echo "First number: $1"
echo "Second number: $2"

sum=`expr $1 + $2`
sub=`expr $1 - $2`
mul=`expr $1 \* $2`
div=`expr $1 / $2`
mod=`expr $1 % $2`

echo "Addition = $sum"
echo "Subtraction = $sub"
echo "Multiplication = $mul"
echo "Division = $div"
echo "Modulus = $mod"
```

9. Write a shell script to assign a variable using set and print it using echo.

```
set name="Linux"
echo "Operating System is: $name"
```

10 Write a shell script to declare two variables and display them using echo.

```
course="BCA"
year=3
echo "Course: $course"
echo "Year: $year"
```

11. Write a shell script to demonstrate unset command.

```
city="Rajkot"
echo "Before unset: $city"

unset city
echo "After unset: $city"
```

12. Write a shell script to assign a subject, print it, then remove it using unset.

```
set subject="Operating System"
echo "Subject is: $subject"
```

```
unset subject
echo "After unset, subject is: $subject"
```

13. Write a shell script where you take input from user, store it in a variable, and print it using echo

```
echo "Enter your name: "
read username
echo "Hello $username, welcome to Shell Scripting!"
```

14.Create two files file1.sh and file2.sh. In file1.sh, define two variables: one with export and one without. Call file2.sh from file1.sh

1. vi file1.sh

```
myvar="Hello"
export exported_var="Exported Hello"
sh file2.sh
```

2. Vi file2.sh

```
echo "Inside file2.sh"
echo "myvar = $myvar"
echo "exported_var = $exported_var"
```

3. run:

```
Chmod 777 file1.sh
Chmod 777 file2.sh
Sh file1.sh
```